

Personalized Netflix Recommendation System

EDA + Collaborative Filtering + Matrix Factorization + MAP@10 Evaluation

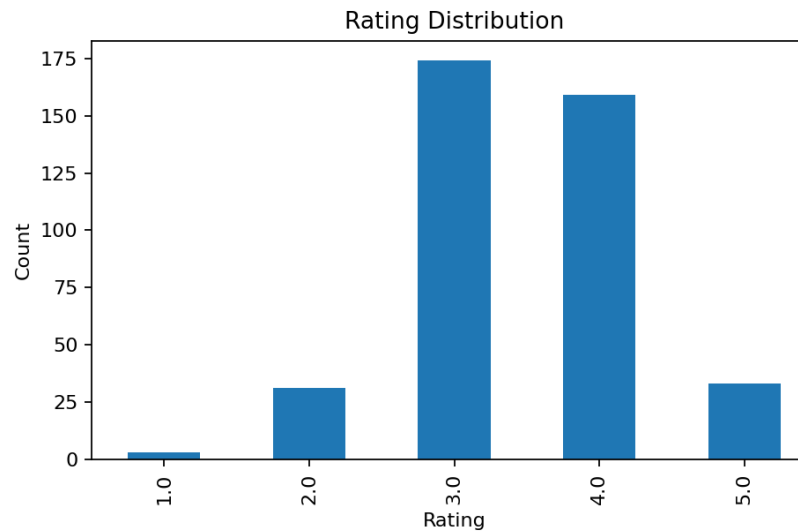
Complete reproducible project with dashboard, report, and recommendation examples

Problem Overview

- Learn user preferences from historical user-movie ratings.
- Predict unseen ratings and generate personalized Top-K recommendations.
- Identify similar content and make recommendations explainable.
- Evaluate both rating accuracy and ranking quality.

EDA Findings

- Demo ratings: 400; users: 40; movies: 18.
- Mean rating: 3.47; sparsity: 44.44%.
- Ratings are sparse, so pure popularity is not enough for personalization.



Model Design

- Bias baseline: global mean + user/movie bias for a hard sanity-check benchmark.
- Item-based collaborative filtering: recommends movies similar to those the user liked.
- Matrix factorization: learns latent user/movie vectors for hidden preference patterns.
- Temporal split prevents future information from leaking into training.

Evaluation Metrics

- RMSE: measures predicted rating error.
- MAE: interpretable average absolute rating error.
- MAP@10: measures ranking quality for the Top-10 list.
- Relevant movie definition: actual rating ≥ 3.5 .

Experimental Results

model	RMSE	MAE	MAP@10	HitRate@10
item_item_cf	0.814	0.613	0.419	1.000
matrix_factorization_sgd	0.644	0.520	0.416	1.000
bias_baseline	0.644	0.553	0.377	1.000

- Best demo ranking model: item_item_cf with MAP@10=0.419.

Recommendation Example

Sample Top-5 recommendations

1. The Shawshank Redemption - score 3.99
2. Batman Begins - score 3.91
3. Gladiator - score 3.86
4. The Matrix - score 3.81
5. Ratatouille - score 3.79

- The dashboard lets users inspect recommendation scores and explanations.

Key Insights and Next Steps

- MAP@10 is the metric that makes this a recommender, not just a rating regressor.
- Item-CF is strong for explainability; MF is stronger after larger-scale tuning.
- Next: train larger subsets, blend models, add diversity re-ranking, and implement cold-start logic.
- The project is GitHub-ready: code, docs, report, slides, dashboard, and reproducible commands are included.